Computational Intelligence

Master in Artificial Intelligence

Evolution of Cooperation in the Iterated Prisoner's Dilemma

Exercise 8 Report

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1 Research question and hypothesis

1.1 Research question

Axelrod demonstrated that a genetic algorithm can find cooperative strategies, such as tit-for-tat (TFT), in the iterated Prisoner's Dilemma [1]. However, in reality errors almost never occur. Players may accidentally play an incorrect move. This can cause a long period of punishment between strategies that would normally always cooperate [2].

Will cooperation still develop when a genetic algorithm evolves strategies with some errors? The only variable will be the probability of an error occurring.

1.2 Hypothesis and falsification

Hypothesis A. As action noise increases, the amount of cooperation in the population will decrease. Cooperation is defined as the proportion of cooperative moves in the population. This will be calculated by taking the average of the proportion across generations 401–500 for all runs.

Falsification. If, for either noise condition ($\varepsilon = 0.01$ or 0.05), the 95% confidence interval around the median difference in cooperation includes or is above 0, Hypothesis A will be falsified. We expect the amount of cooperation to decrease as noise increases. If the amount of cooperation with $\varepsilon = 0.05$ is as much or more than with $\varepsilon = 0.01$, Hypothesis A will also be falsified, as this is a stronger, ordered hypothesis.

We might also notice that some noise could encourage more forgiving strategies such as PAVLOV (win-stay, lose-shift) [2]. We will look at how the proportion of PAVLOV changes, although this is not a main point.

2 Problem definition

2.1 Payoffs and action noise

There are two players playing for 150 rounds. In each round both players cooperate (C) or defect (D). The payoffs are taken from the exercise [4]. The payoffs are $T = 5$ (defect against cooperator), $R = 3$ (mutual cooperation), $P = 1$ (mutual defection), and $S = 0$ (cooperate against defector). These values mean that mutual cooperation is the optimal collective outcome as $T > R > P > S$ and $2R > T + S$. However, it is always tempting to defect.

Action noise is introduced. When playing a move, both players select an action based on their strategy. With probability ε this action is changed. The errors of both players are independent. Both players know what action was actually played. This determines the payoff and the next move. We look at $\varepsilon = 0$, 0.01, and 0.05. With $\varepsilon = 0.01$, a pair of players will make approximately 3 mistakes in a 150-round match. With $\varepsilon = 0.05$, they will make approximately 15 mistakes.

2.2 Memory-one strategies

All strategies only depend on the last round. There are five bits representing what the strategy will do. The first bit represents the initial action. The next four bits represent what the strategy will do in response to the outcomes of the last round: CC, CD, DC, or DD. The first action in an outcome represents the player's own action. So CD means "I cooperated, the opponent defected." A 0 means cooperate and a 1 means defect. There are

$2^5 = 32$ strategies. The strategies are numbered 0–31 by interpreting the bits as a binary number. Table 1 shows the five strategies we examine.

Table 1: The five named strategies. Bits are [first action, after CC, after CD, after DC, after DD].

ID	Bits	Name	Behaviour
0	00000	ALLC	Always cooperates.
5	00101	TFT	Starts with C, then copies the opponent's last action.
6	00110	PAVLOV	Starts with C. Keeps its action after R or T, switches after S or P.
7	00111	GRIM	Starts with C. Cooperates only after mutual cooperation.
31	11111	ALLD	Always defects.

GRIM here is the memory-one version. Without noise it never forgives. Because it only looks one round back, it can return to cooperation if both players happen to cooperate in the same round by mistake. Under noise this has a chance of about ε^2 per round, so our GRIM is slightly less harsh than the classical grim trigger.

3 Methods

3.1 Round robin and expected-payoff matrices

Each agent plays against each other agent once, for 150 rounds. The agent's fitness is based on the average payoff it received per round across the 99 matches.

If we simulated all the matches, this could take some time and would introduce randomness into fitness. However, as there are only two memory-one strategies, we can create a small Markov chain with the four possible outcomes (CC, CD, DC, DD). We calculate the expected outcome exactly based on the first two bits and the noise. After calculating the probabilities of each outcome in round one, we calculate subsequent rounds using a 4×4 transition matrix created from the response bits. We sum all of the outcome probabilities across the 150 rounds to find expected payoffs and the expected percentage of cooperative actions between the pair.

We do this for all 32×32 pairs, giving us two matrices for each noise level. $A[i, j]$ is the expected payoff per round between strategy i and strategy j. $H[i, j]$ is the expected percentage of cooperative actions during the match. This is calculated based on whether the outcome was CC (1), CD/DC (0.5), or DD (0). For a population of n strategies, the fitness of strategy i is calculated as

$$\frac{A[i] \cdot n - A[i, i]}{99}.$$

This removes self-play. We use H in the same way to find population cooperation. We have tested these matrices using random simulation and they agree.

3.2 Genetic algorithm

The genetic algorithm was created using DEAP [3, 5]. Each individual is a list of five bits. In each generation we evaluate everyone against the current population. The top

two individuals are copied exactly. We select 98 parents using tournament selection and perform uniform crossover and bit-flip mutation to create new individuals. Each generation we recalculate fitness as the opponents have changed. Table 2 shows the parameters used. These are the same for all conditions; only the noise differs.

Table 2: Genetic algorithm settings, identical for all conditions.

Setting	Value
Population size	100
Generations	500
Rounds per match	150
Crossover probability per pair	0.7
Uniform crossover, swap chance per bit	0.5
Mutation, flip chance per bit	0.01
Tournament size	3
Elites copied unchanged	2
Final window	Generations 401 to 500

3.3 Noise levels and baseline controls

The noise-free case ($\varepsilon = 0$) can be considered as a baseline. This follows Axelrod's scenario, where cooperation is expected to be successful. All other conditions are unchanged, meaning any differences in results must be due to the changes in noise. Another baseline is the payoff between the five strategies mentioned in Section 5.1. This remains unchanged.

3.4 Pilot studies, independence, and seed

Thirty independent runs were completed for each noise level, meaning a total of 90. Each run has an independent seed given by $21261112 + 10000c + r$, where c is the condition index (0, 1, or 2 for noise 0, 0.01, and 0.05) and r is the run index (0–29). All runs were independent of each other, and therefore the three sets of data were independent samples.

We decided to complete a pilot study using the first 10 runs of each condition. This was used as a retrospective pilot study, meaning that there was no additional evidence from it as these were not extra runs. We found that the variation between runs was quite small compared with the difference between the conditions. The interquartile range for cooperation was 0.2, 4.4, and 18.8 between each ε value. However, the medians between conditions varied by between 37 and 88. Even when using only the 10 pilot runs, all three confidence intervals were very far from zero. We therefore chose to complete 30 runs to reduce the confidence intervals. This was mainly to decrease the confidence interval for the comparison between 0.01 and 0.05, which had the most variation.

3.5 Measures, effects, and confidence intervals

Our main measure was cooperation, which is the proportion of cooperative actions out of all matches within the population. This was calculated using an average of 100 generations (401–500). Averaging over 100 generations helps to prevent short-term spikes due to mutations. Our secondary measure was the average payoff per round, also calculated for each run.

For each pair of noise levels we calculated the difference in medians between the noisier and less noisy conditions. This was calculated with a 95% percentile bootstrap confidence interval using 5,000 resamples of the runs. We also conducted a two-sided Mann–Whitney U test. This was chosen due to the independent nature of our samples and because we did not assume the data to follow a normal distribution. Finally, we calculated the Vargha–Delaney effect size $\hat{A}_{12}$ [6]. This tells us the probability that a randomly selected run from the noisier condition will have a higher score than a run from the other condition. Ties were counted as half. A value of 0.5 indicates no effect, whilst a value of 0 indicates all noisy runs had a lower score. Due to having three pairs of noise levels and two measures, six tests were conducted. To account for this we adjusted the p-values using the Holm method [7].

4 Results

4.1 Payoff and cooperation over generations

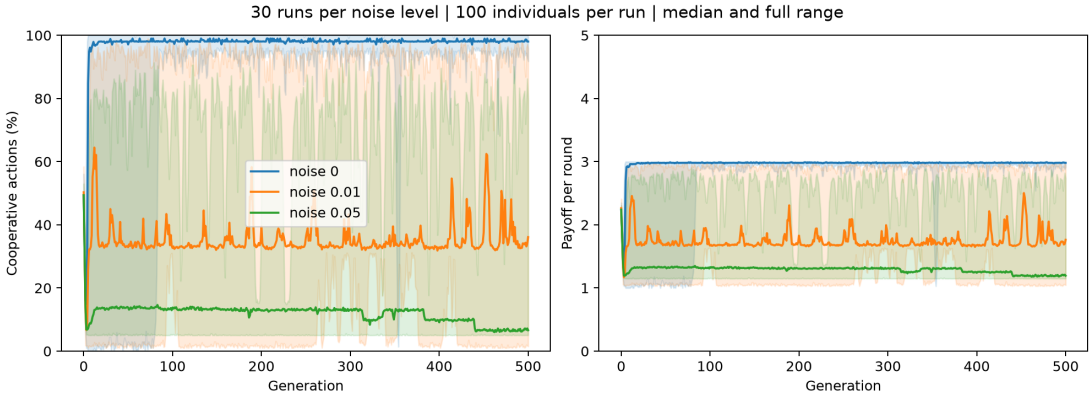


Figure 1: Cooperation (left) and mean payoff per round (right) in each generation. Lines show the median of 30 runs; shaded areas show the full range.

The three scenarios evolve very differently (Figure 1). All begin with around 50% cooperation, as expected for random strategies, but quickly diverge after around 10 generations.

In the non-noisy scenario cooperation rises to around 98% and remains constant at this value throughout. The payoff is just under 3, which is the value for mutual cooperation. However, for $\varepsilon = 0.01$ the level of cooperation dips to around 35% on average in each generation, although there are frequently brief bursts of 50% or 60%. For $\varepsilon = 0.05$ cooperation falls even further to around 13% and gradually decreases further so that by the end it is only around 7%. Note that the shaded regions show that occasional high levels of cooperation do occur during individual noisy runs.

4.2 Strategies and stability

Figure 2 shows how the strategies evolve. When $\varepsilon = 0$, TFT is dominant with a final population around 72% and is the most prevalent strategy over all 30 runs. The remaining strategies are mostly strategy 00100 (around 17%). This strategy differs from TFT in the outcome after DD. However, with no noise DD is never played, so they receive the same payoffs and undergo random drift.

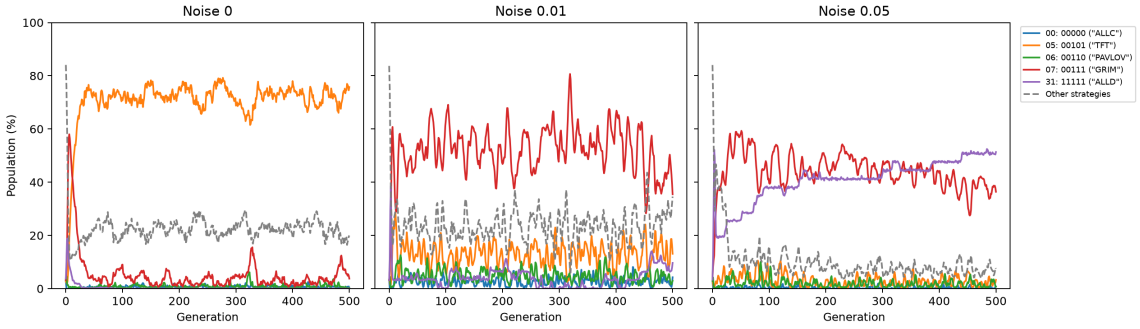


Figure 2: Share of the five named strategies over time, averaged over 30 runs per noise level.

When $\varepsilon = 0.01$, GRIM becomes dominant with a population of around 49%. GRIM dominates in 27/30 runs. TFT is still prevalent with a population of around 14%. GRIM varies in population between around 35–80%.

When $\varepsilon = 0.05$, the population is divided, with ALLD (50%) and GRIM (38%). ALLD dominates in 16 of the runs and GRIM dominates in 14. The runs where ALLD dominates have an equilibrium level of around 6% cooperation and the runs where GRIM dominates have an equilibrium level of around 22%. This can be seen in Figure 3 with two distinct groups when $\varepsilon = 0.05$.

PAVLOV never becomes prevalent. The average population remains below 5% for all levels of noise. PAVLOV never dominates any runs.

4.3 Final-window comparisons and uncertainty

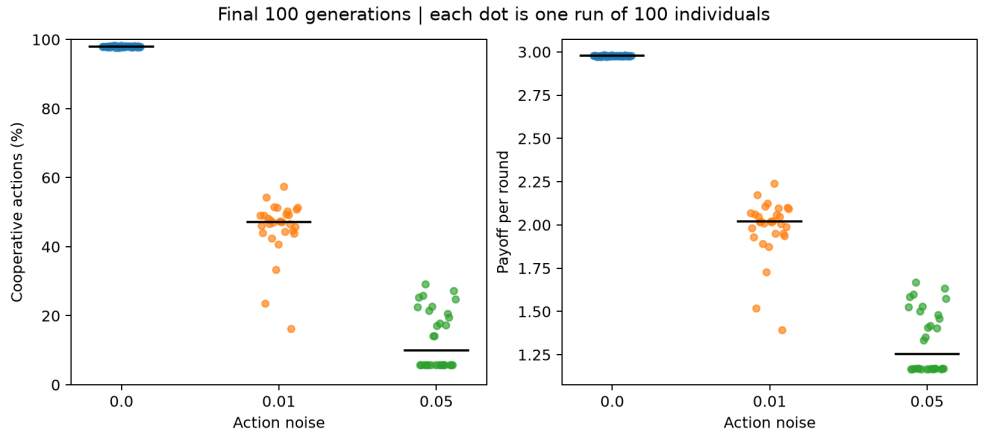


Figure 3: Average over generations 401 to 500 for each run. Each dot is one run of 100 individuals, and the black bar marks the median.

The final results of the comparison can be seen in Table 3 and Figure 3. The median level of cooperation was 98.0% when $\varepsilon = 0$, but fell to 47.2% when there was some noise added ($\varepsilon = 0.01$), and to 10.0% at $\varepsilon = 0.05$. This is a drop of 50.8 percentage points from $\varepsilon = 0$ to $\varepsilon = 0.01$, and an overall drop of 88.0 percentage points from $\varepsilon = 0$ to $\varepsilon = 0.05$. There is no confidence interval that is near zero, and all Holm-corrected p-values are below 10^{-9}.

There are also very large effect sizes. For both comparisons to the baseline, $\hat{A}_{12} = 0$,

Table 3: Differences between noise levels over generations 401 to 500 (30 runs per group). Differences are noisier minus less noisy. Cooperation is in percentage points, payoff in points per round.

Comparison	Measure	Median diff.	95% CI	Holm p	$\hat{A}_{12}$
0.01 vs 0	Cooperation	−50.8	−52.3 to −48.9	$< 10^{-9}$	0.00
0.05 vs 0	Cooperation	−88.0	−92.3 to −79.2	$< 10^{-9}$	0.00
0.05 vs 0.01	Cooperation	−37.2	−42.8 to −28.1	$< 10^{-9}$	0.02
0.01 vs 0	Payoff	−0.96	−0.99 to −0.92	$< 10^{-9}$	0.00
0.05 vs 0	Payoff	−1.72	−1.81 to −1.54	$< 10^{-9}$	0.00
0.05 vs 0.01	Payoff	−0.77	−0.88 to −0.57	$< 10^{-9}$	0.02

meaning that all runs with some noise had less cooperation than all runs without any noise. For the comparison between $\varepsilon = 0.01$ and $\varepsilon = 0.05$, $\hat{A}_{12} = 0.02$, meaning that almost all runs with $\varepsilon = 0.05$ have less cooperation than almost all runs with $\varepsilon = 0.01$. The payoffs show the same trend.

As the refuting outcome did not happen, Hypothesis A has been supported. As expected, cooperation decreases in the order predicted.

5 Mechanism and discussion

5.1 Classical strategy payoff matrices

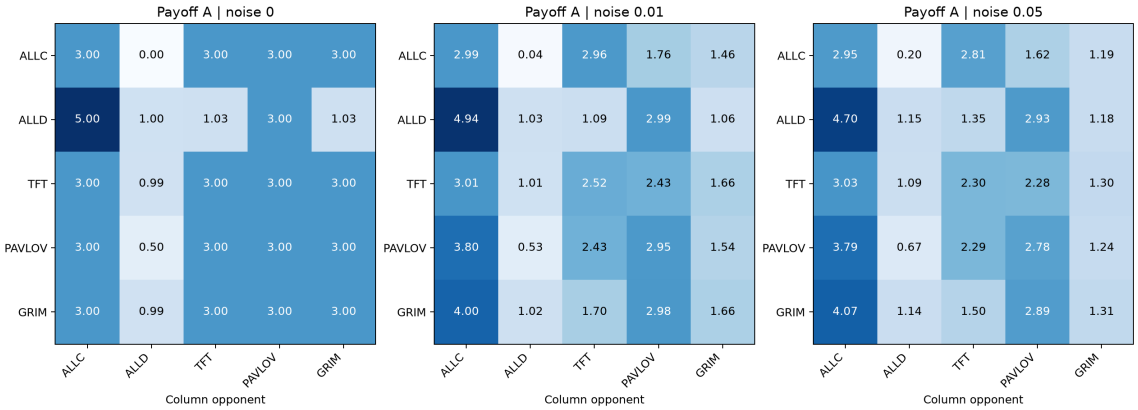


Figure 4: Expected payoff per round of the row strategy against the column strategy, from matrix A.

The expected payoffs for all five strategies are shown in Figure 4. With no noise all four nice strategies have a payoff of 3 against each other, making them indistinguishable under evolution. However, ALLD has a low payoff of only about 1 against both TFT and GRIM, thus resulting in extinction.

With noise, the expected payoff of TFT against itself becomes less than 3. The payoff against itself drops to 2.52 and 2.30 for $\varepsilon = 0.01$ and 0.05, respectively. This occurs because one error results in an endless cycle of reciprocation that TFT cannot escape. The payoff for PAVLOV, however, remains high at 2.95 and 2.78 ($\varepsilon = 0.01$ and 0.05). This is because PAVLOV will rectify an error in two rounds. Unforgiving strategies receive higher payoffs against forgiving strategies. GRIM received a payoff of 4.00 against ALLC and 2.98 against

PAVLOV ($\varepsilon = 0.01$). However, TFT had a lower payoff of only 3.01 and 2.43. ALLD also had a higher payoff against PAVLOV of 2.99.

5.2 Evidence for the proposed mechanism

We propose that without any noise all the nice strategies are equivalent, giving a safe environment to cooperate. Once errors occur, the strategies start to become distinguishable based on how they respond to forgiving opponents. Exploitative strategies benefit from this. Due to mutation, more cooperative strategies such as ALLC and PAVLOV will be introduced into the population. This rewards the tough strategies (e.g., GRIM and ALLD), causing cooperation to decrease.

To investigate this, we calculated the expected payoff for each strategy against the evolved population. For each noise value we used the average strategy distribution between generations 401–500 for all 30 runs. Using this we calculated the expected fitness of each strategy using matrix A (Table 4).

Table 4: Expected fitness of each named strategy against the evolved population (mean strategy mix in generations 401 to 500). The highest value per column is in bold.

Strategy	$\varepsilon = 0$	$\varepsilon = 0.01$	$\varepsilon = 0.05$
ALLC	2.98	1.75	0.75
TFT	**2.99**	1.98	1.29
PAVLOV	2.98	2.02	1.06
GRIM	2.97	**2.04**	**1.34**
ALLD	1.55	1.75	1.30

These fitness values back up the mechanisms. At $\varepsilon = 0$, all four nice strategies have very similar fitnesses (within 0.02). This is the neutral regime where both TFT and 00100 can drift. At $\varepsilon = 0.01$, GRIM has the highest fitness (2.04), followed by TFT (1.98). PAVLOV has a similar average fitness (2.02), but it works best in a population with lots of other PAVLOV strategies. In a population with all GRIM strategies, PAVLOV has a low fitness (1.54), while GRIM has a higher fitness (1.66). This means a minority of PAVLOV cannot increase in population. When ε increases to 0.05, GRIM (1.34), TFT (1.29), and ALLD (1.30) have similar fitness values. Such small differences mean the strategy will be determined by chance. This can be seen in the almost equal proportions between the ALLD and GRIM runs.

The winning strategy is also the least cooperative. In the evolved population at $\varepsilon = 0.01$, TFT had 46% cooperative interactions while GRIM had 35%. This helps explain why cooperation decreased when changing from TFT to GRIM.

This also answers this side question. We found no evidence that PAVLOV is favoured by moderate levels of noise. While it is a good cooperator with itself, it is exploited by both GRIM and ALLD strategies, meaning it never becomes prevalent.

6 Limitations

6.1 Assumptions of the model

This conclusion only applies to this model, as there were some assumptions made.

Only deterministic, memory-one strategies were possible. Strategies with longer memory or random responses (such as generous tit-for-tat) would perform much better with noise. However, these strategies were not in our search space.

We only tested at three noise levels. We know that at some point between 0 and 0.01, cooperation breaks down, but we do not know where.

The mutation rate was fixed at 0.01 per bit. This means there will always be cooperative mutants. Changing the mutation rate, population, or selection pressure may change the results. We did not test this.

The number of runs was limited. At $\varepsilon = 0.05$ there was still a downward trend in cooperation at generation 500. This means that the final window may not be a true end state.

The payoffs are exact. This removes any randomness in fitness due to noisy interactions.

The strategies in Table 4 are an average of the proportion of strategies throughout the runs. An individual run can look very different, especially at $\varepsilon = 0.05$.

7 Conclusions

Cooperation evolved in all runs with no noise. However, any small amount of noise in actions prevents this from occurring. At $\varepsilon = 0.01$, cooperation decreases to 47% (median), and at $\varepsilon = 0.05$ it decreases to 10%. There are large differences in all cases. Confidence intervals are far from zero and effect sizes are near their maximums. This supports Hypothesis A.

This is because a different strategy is winning. The noise causes TFT players to punish each other, and strategies that can exploit the forgiving mutants are rewarded. At $\varepsilon = 0.01$, GRIM replaces TFT and at $\varepsilon = 0.05$ both GRIM and ALLD take over.

8 Reproducibility

All of the code and results can be found in our project folder. README.txt explains how they were set up. We used Python 3.12.14, NumPy 2.5.4, DEAP 1.4.4, Matplotlib 3.11.2, and SciPy 1.18.1. All packages are fixed using requirements.txt. To reproduce all 90 runs, CSV files, and figures, run the command `python run_experiment.py -runs 30`. Seeds are described in Section 3.4. The bootstrap was run using seed 22161112 and 5,000 resamples. The method can be seen in detail in the notebook `exercise_8.ipynb`. One population per noise level can be seen live. As fitness is based on exact expected payoffs, results only depend on the seeds of the genetic algorithm.

9 Generative AI declaration

We used Claude Code (Anthropic, 2026) [8] in this project. It looked through our code and gave suggestions for improvements. These were mostly related to the statistical analysis. It also helped with fixing some punctuation errors and some inconsistencies in our report. All suggestions were checked and all numbers in this report have been checked against our result files. Everything in this report is our responsibility. Finally, we used it to convert the report to LaTeX for the final PDF.

References

[1] Axelrod, R. (1987). The evolution of strategies in the iterated prisoner's dilemma. In L. Davis (Ed.), *Genetic Algorithms and Simulated Annealing* (pp. 32–41). Pitman. Author-hosted adapted chapter: https://websites.umich.edu/~axe/research/Evolving.pdf.

[2] Nowak, M., and Sigmund, K. (1993). A strategy of win-stay, lose-shift that outperforms tit-for-tat in the Prisoner's Dilemma game. *Nature*, 364, 56–58. https://doi.org/10.1038/364056a0.

[3] Fortin, F.-A., De Rainville, F.-M., Gardner, M.-A., Parizeau, M., and Gagné, C. (2012). DEAP: Evolutionary Algorithms Made Easy. *Journal of Machine Learning Research*, 13, 2171–2175. https://jmlr.org/papers/v13/fortin12a.html.

[4] Belanche, L. A. (2026). Evolutionary Computation practical work. Computational Intelligence, Master in Artificial Intelligence. Exercise 8 and general instructions. Course handout supplied as `practical_exercises.pdf`.

[5] DEAP developers. Evolutionary Tools and Algorithms. Documentation. https://deap.readthedocs.io/en/master/api/tools.html and https://deap.readthedocs.io/en/master/api/algo.html (accessed 7 November 2026).

[6] Vargha, A., and Delaney, H. D. (2000). A critique and improvement of the CL common language effect size statistics of McGraw and Wong. *Journal of Educational and Behavioral Statistics*, 25(2), 101–132.

[7] Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6, 65–70.

[8] Anthropic. (2026). Claude Code (Claude Opus 5.5 version) [Large language model]. https://claude.com/claude-code.